Inverse Transition Learning from Expert Demonstrations:
Reproduction and a Joint Reward–Dynamics Extension

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1 Motivation and question

Standard reinforcement learning knows the dynamics T of a Markov decision process and wants the optimal policy π^*. Many real settings invert that assumption: [1] uses an intensive care unit as its running example, where outcomes are observable but how a patient responds to a particular dose is not, and what we see is a batch of clinician decisions that are close to optimal but not exactly optimal.

My broader interest sits downstream of this in personalized wellness technology: the wearable stack (Apple Watch, Oura, WHOOP, continuous glucose monitors). Most consumer pipelines fit a supervised model from user action to outcome, treating every body as a shared-dynamics black box and prescribing one-size-fits-all interventions. Inverse Transition Learning (ITL) takes a different posture: the user is a near-optimal decision-maker whose actions encode information about their own physiology, and that signal is exactly what maximum-likelihood-over-counts throws away. The semester's technical question: given R and a batch of ε-optimal expert transitions $D = \{(s, a, s')\}$, can we recover T^* in a way that respects the optimality information in the actions?

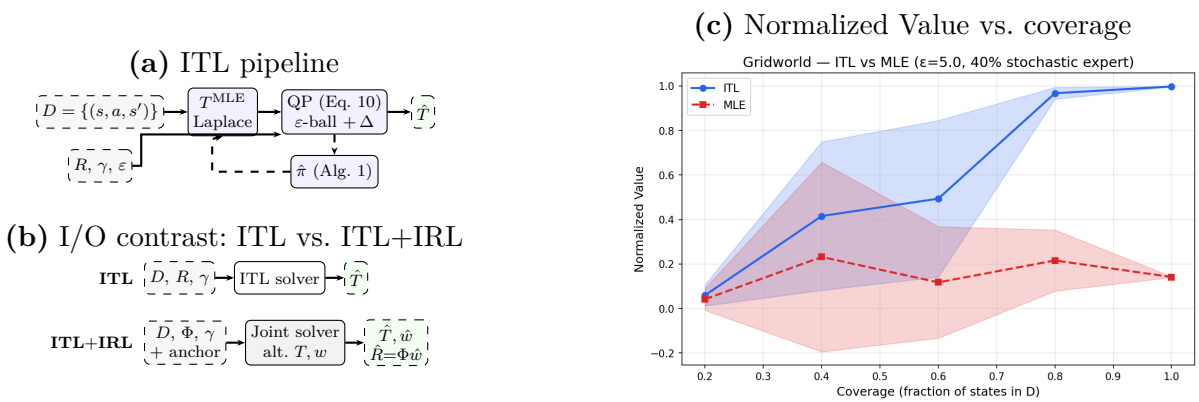

Figure 1: (a) ITL pipeline: the QP projects T toward T^{MLE} subject to ε-optimality of observed actions; $\hat{\pi}$ and the QP alternate until convergence. (b) The proposed extension lifts R into a decision variable $R = \Phi w$ and recovers $(\hat{T}, \hat{w})$ jointly. (c) On Gridworld, ITL plateaus near 1.0 at full coverage and dominates MLE at every coverage level (5 seeds; bands ± 1 std.).

2 Method and tools

An action a is ε-optimal at state s when $\max_{a'} Q^*(s, a'; T) - Q^*(s, a; T) \leq \varepsilon$. Observing the expert choose a at s yields an inequality on Q^*. Writing the policy-evaluation inverse $v^\pi = (I - \gamma P^\pi)^{-1} r^\pi$ and substituting the Laplace-smoothed MLE T^{MLE} inside it linearizes the constraint in T, giving

$$\min_T \ \frac{1}{2} \sum_{s,a,s'} N_{s,a,s'} \left(T_{s,a,s'} - T^{\mathrm{MLE}}_{s,a,s'}\right)^2 \quad \text{s.t.} \quad \varepsilon\text{-ball on } (s,a) \in D, \text{ simplex on } T.$$

The solver alternates between $\hat{\pi}$ (uniform over observed actions; uniform over all actions at unobserved states) and the QP, terminating when the ε-ball property holds at every observed state (Fig. 1a). The convex pieces map onto Krause and Hübotter [2, Ch. 10–11]: Bellman optimality, the linear policy-evaluation system, and the Dirichlet–Categorical conjugacy that makes the Laplace-smoothed MLE a posterior mean rather than a frequentist estimate. Implementation: CVXPY [5] with an OSQP–SCS–CLARABEL fallback chain.

3 Simulation: coverage sweep on Gridworld

I reproduced the paper's central claim (their Figure 2) on a 5×5 Gridworld with soft walls (penalty -5), slip 0.2, γ=0.95, ε=5.0, 40%-stochastic expert. *Coverage* is the fraction of states appearing in D; for each observed (s, a) I sampled 10 transitions. At full coverage, ITL achieves Normalized Value 0.998 ± 0.001 vs. MLE's 0.142 ± 0.002, with 0 vs. 7.20 constraint violations and ε-matching 1.000 vs. 0.712 (5 seeds; Fig. 1c). MLE under-counts visited next-state cells under a stochastic expert, so its greedy policy walks into soft walls; ITL forces the expert's observed choices to remain ε-optimal under $\hat{T}$. Shape matches the paper's Figure 2; absolute scale runs $\sim 1.5\times$ larger than Table 4, almost certainly because the soft-wall layout is not published in detail. The structural pattern (ITL $\geq$ MLE on every metric, ε-matching $\to 1.0$ at full coverage per Theorem 1) holds on both Gridworld and RandomWorld.

4 Open question: joint reward and dynamics recovery

The paper assumes R is known. In MIMIC-IV that is fragile: the survival proxy reward (± 15 for discharge or mortality, after [4]) is hand-engineered. The extension I have been prototyping, named in the paper's future-work section, lifts R into a decision variable $R(s, a) = \Phi(s, a)^\top w$ and solves for (T, w) jointly (Fig. 1b). The ε-ball constraint becomes

$$\big(\Phi(s,a) - \Phi(s,a')\big)^\top w + \gamma\,(T_{s,a} - T_{s,a'})^\top v_{\text{lin}} \;\geq\; \varepsilon,$$

bilinear in (T, w) but a convex QP at each step of the alternation (hold w, solve for T; hold T, solve for w). An L1 penalty on w plus an anchor at one (s, a) pin the additive ambiguity. The prototype converges to a fixed point in two outer iterations on both the 3-state corridor and the 5×5 Gridworld. On Gridworld with one-hot (s, a) features ($d = 100$), goal-anchored at $R(s^*, 0) = 10$, $\lambda_{L1} = 0.01$, deterministic expert: ITL+IRL recovers the optimal policy exactly (best-action 1.000 vs. ITL with known R 1.000 vs. MLE 0.600); T MSE 0.4873 vs. 0.4929 (ITL) vs. 0.5154 (MLE). *Honest caveat:* this gridworld is goal-dominated — pairing T_{MLE} with a trivial reward R=$10 \cdot \mathbf{1}[s{=}s^*]$ recovers π^* at the same rate as pairing it with R^*, so a 1.000 best-action match is necessary but not sufficient evidence that the IRL step is doing real work. A two-goal benchmark (R_A=5, R_B=10) and a head-to-head against MaxEnt-IRL [3] (wired in via L-BFGS-B with a null-space anchor) are the next falsifiers before May 31.

Implications for personalized wellness. The reproduction confirms that under near-optimal behavior, individual-level dynamics are recoverable from sparse demonstrations in a way supervised models miss. The joint reward extension matters more for wearables than the ICU: sleep-quality and recovery scores are noisy user proxies, so jointly inferring R with T removes the most fragile assumption in any wearables pipeline. The continuous-state lift is the gating step before this transfers to raw sensor streams.

Code. Reproduction codebase (Python 3.10, CVXPY, MIT) at `https://github.com/biohackingmathematician/itl`, with inline cross-references to paper equations and textbook chapters and 8 pytest smoke tests covering the corridor value function, ITL convergence, BITL constraint regression, and the goal-domination invariant. A Colab re-running the corridor and gridworld experiments will be linked from the README.

References

[1] Benac, L., et al. (2024). Inverse Transition Learning. *AAAI 2025* (preprint).

[2] Krause, A., and Hübotter, J. (2025). *Probabilistic Artificial Intelligence*. Cambridge Univ. Press, Ch. 10–12.

[3] Ziebart, B. D., et al. (2008). Maximum Entropy Inverse Reinforcement Learning. *Proc. AAAI*.

[4] Komorowski, M., et al. (2018). The AI Clinician learns optimal treatment strategies for sepsis in ICU. *Nature Medicine* 24, 1716–1720.

[5] Diamond, S., and Boyd, S. (2016). CVXPY: A Python-Embedded Modeling Language for Convex Optimization. *JMLR* 17(83), 1–5.